Sebastián Forero Duque

Systems and Computer Engineer | Data Science | Machine Learning | Process Automation (IBM Cloud Pak)
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PROFESSIONAL SUMMARY

Systems and Computer Engineer from Universidad Pontificia Bolivariana (UPB), currently pursuing a Master's in Data Science at the same university since August 2026. Experience in data science: predictive and descriptive modeling with Machine Learning (Scikit-learn, TensorFlow, Keras), ETL pipelines, data analysis and visualization (Power BI, Tableau, Streamlit) and relational databases. I complement this profile with hands-on experience in business process automation: during my internship at XM S.A. E.S.P. (ISA Group) I automated data workflows and enterprise architecture processes with IBM Cloud Pak for Business Automation (CP4BA) and IBM Business Automation Studio. English level C1 certified by EF Education First and an overall band score of 7.5 in IELTS General Training.

PROFESSIONAL EXPERIENCE

IT Intern, Process Automation | XM S.A. E.S.P. (ISA Group)
Medellín, Colombia | January to July 2026

Developed the end-to-end automation of the Architecture Technical Committee process using IBM Business Automation Studio (BAW) within IBM Cloud Pak for Business Automation (CP4BA), covering the full session lifecycle, from the initial request to commitment follow-up.

- Designed and built dynamic forms and business logic in BAW to manage the complete architecture committee workflow.
- Developed data handling and integration logic to store requests, decisions and commitments in a structured way.
- Connected the platform to internal APIs to trigger automatic notifications and keep stakeholders informed in real time.
- Supported and optimized the meeting room booking process for architecture sessions, reducing manual coordination effort.
- Built Power BI dashboards from session metrics, giving visibility into meeting volume, scheduling efficiency and commitment tracking for management decision making.

RELEVANT PROJECTS

Air Quality Dashboard: ETL and Data Visualization | UPB / SIATA
Medellín, Colombia | 2025

- Built a complete ETL pipeline to extract, transform and load environmental sensor data from SIATA, converting PM2.5 measurements into Air Quality Index (AQI) values according to EPA standards.
- Developed an interactive geospatial dashboard in Streamlit with Pandas, Matplotlib and map layers, with real-time filters by zone and time range.
- Covered the full data engineering and data science cycle: ingestion, transformation, storage and visualization.

Emotion Detection Model: Predictive Modeling with ML | UPB
Medellín, Colombia | 2024

- Developed a deep learning model with Convolutional Neural Networks (CNN) in TensorFlow and Keras to classify emotions in real time from video.
- Tuned hyperparameters and applied data augmentation to reduce the false positive rate and improve model generalization.
- Performed descriptive analysis of the training data distribution to identify class imbalance and improve model fairness.

Supplier Evaluation Platform: Backend | UPB / CTIC
Medellín, Colombia | 2024

- Led a development team that built a full stack application with Java (Spring Boot), MySQL and Angular to automate supplier evaluation under ICONTEC standards.
- Designed the relational database schema and stored procedures to ensure data integrity and auditability.

REST API for Chemical Compound Management | UPB
Medellín, Colombia | 2024

- Built a RESTful API with Spring Boot and PostgreSQL, with stored procedures and complex relational queries to manage chemical compounds and elements.

TECHNICAL SKILLS

Data Science and Machine Learning: Scikit-learn, TensorFlow, Keras, predictive modeling, descriptive modeling, CNN, neural networks, Transformers, NLP (basic), model training and evaluation, hyperparameter tuning

Data Engineering and ETL: Pandas, ETL pipeline design, data ingestion, data transformation, data validation, SQL, PostgreSQL, MySQL, MongoDB, NoSQL, data modeling, data warehousing

Visualization and Analytics: Power BI, Tableau, Streamlit, Matplotlib, Seaborn, Pandas, Jupyter Notebook

Programming Languages: Python, Java, JavaScript, C#

Cloud and Infrastructure: AWS (EC2, S3), Docker, IBM Cloud Pak for Business Automation (CP4BA), IBM Business Automation Studio, IBM Cloud Pak for Data

Backend and APIs: Spring Boot, REST API development and consumption

Automation and Architecture: BPMN, IBM Business Automation Workflow (BAW), ArchiMate, UML

Applied AI: Model Context Protocol (MCP), Claude Code assisted development, AI agent integration in workflows

Other Tools: Git, GitHub, Postman, n8n

EDUCATION

B.Sc. in Systems and Computer Engineering | Universidad Pontificia Bolivariana (UPB)
Medellín, Colombia | 2022 to 2026 | Degree obtained

Master's in Data Science | Universidad Pontificia Bolivariana (UPB)
Medellín, Colombia | August 2026 to present

English C1 Certification | EF Education First
Brighton, England | 2024

IELTS General Training | Overall band score: 7.5
2026

High School Diploma with Honors | Colegio de la UPB
Medellín, Colombia | 2010 to 2021

CERTIFICATIONS

- IBM Cloud Pak for Data 5.0: Technical Essentials | IBM | Jun. 2026
- Model Context Protocol: Advanced Topics | Anthropic | Mar. 2026 | ID: czud6fzdsbxq
- Claude 101 | Anthropic | Mar. 2026
- Claude Code in Action | Anthropic | Jan. 2026 | ID: qu24ivpdi4e8
- Python | freeCodeCamp | Aug. 2026 | ID: fcc-861b3ac9-8d2f-43b0-8171-f03b782ce1cd-pyv9

LANGUAGES

Spanish: native | English: C1 (EF Education First, 2024) | IELTS General Training 7.5 (2026)